

**METRO
MANAGEMENT
SYSTEM**

OVERVIEW

- Metro Management System is a software platform for managing metro services.
- Provides login access to both users and admin.
- Facilitates ticket booking, route tracking, and administrative controls.

USER AND ADMIN LOGIN

- This allows both Admin and User to log in by entering their credentials. It checks the entered ID and passcode against two different tables in the database: **admin** and **credentials**.

QUERIES FOR USER AND ADMIN LOGIN

FOR USER

```
SELECT * FROM credentials
```

```
WHERE user_id = %s AND passcode = %s;
```

FOR ADMIN

```
SELECT * FROM admin
```

```
WHERE user_name = %s AND passcode = %s;
```

TRAIN SCHEDULE

Displays detailed metro schedules including arrival and departure times. Also shows platform number and station information for each metro.

Associated Queries for this :

```
SELECT
ms.metro_id,
ms.route_id,
s.st_name AS station_name, s.line_color,
ms.arrival_time, ms.departure_time,
ms.platform,
ms.direction
FROM
    metro_schedule ms
JOIN
    station s ON ms.st_code = s.st_code
WHERE
    (Conditions added dynamically based on filters)
ORDER BY
ms.metro_id,
    FIELD(ms.direction, 'Forward', 'Reverse'),
    Ms.arrival_time;
```

CHECK METRO ROUTES

Associated Queries for this :

Here we can find routes between stations. Also we can find the schedules of metros which run on the route given.

```
SELECT
    ms1.metro_id,
    m.metro_name,
    ms1.st_code AS start_station,
    ms1.arrival_time AS start_time,
    ms2.st_code AS end_station,
    ms2.arrival_time AS end_time
FROM
    metro_schedule ms1
JOIN
    metro_schedule ms2 ON ms1.metro_id = ms2.metro_id
JOIN
    metro m ON ms1.metro_id = m.metro_id
WHERE
    ms1.st_code = (
        SELECT st_code
        FROM station
        WHERE st_name = '<source>'
        LIMIT 1
    )
    AND ms2.st_code = (
        SELECT st_code
        FROM station WHERE st_name = '<destination>'
        LIMIT 1
    )
    AND ms1.arrival_time < ms2.arrival_time
ORDER BY
    ms1.arrival_time
```

METRO SCHEDULE MANAGEMENT

Admin can update the arrival or departure times of all the metros.

Associated Queries for this:

View & Search Schedule Entries

```
SELECT ms.schedule_id,  
ms.metro_id, m.metro_name,  
ms.st_code, s.st_name,  
    ms.arrival_time,  
ms.departure_time, ms.platform,  
ms.direction  
FROM metro_schedule ms  
JOIN metro m ON ms.metro_id =  
m.metro_id  
JOIN station s ON ms.st_code =  
s.st_code  
WHERE ms.metro_id LIKE  
%search%  
ORDER BY ms.metro_id,  
ms.arrival_time
```

Update a Schedule Entry

```
UPDATE metro_schedule  
SET arrival_time = %s,  
    departure_time = %s,  
    platform = %s  
WHERE schedule_id = %s
```

MANAGING METROS

Admin can add or delete a metro from database.

Associated Queries for this:

Query 1. To add a metro:

```
INSERT INTO metro  
(metro_id, metro_name,  
line_color)  
VALUES (%s, %s, %s)
```

Query 2. To delete metro:

```
DELETE FROM  
metro_schedule  
WHERE metro_id = %s  
AND st_code = %s
```

ADMIN – TICKET HISTORY

- View all ticket bookings:
- - Filter by date, station, line color.
- - Sort by popularity or date.

ADMIN – TRAFFIC ANALYSIS

- Passenger stats by station and weekday.
- Line changes: Calculates passengers changing lines.
- Average passengers per metro based on entry data.

CONCLUSION

- Comprehensive platform for metro management.
- User Features: Booking, schedules, history.
- Admin Features: User control, analytics, route editing.
- Powered by SQL backend and GUI frontend.